# Stage 3 Report: Second Approach, GAN-Based Super-Resolution

**Image Super-Resolution with Deep Generative Models · Computer Vision 2026 · 16 Oct 2026** Team: Iuliia Kurnaeva, Alena Petrenko, Amaliya Kharisova, Dmitrii Kuznetsov

**Repository:** https://github.com/Ephy01/Image-Super-Resolution-with-Deep-Generative-Models

## 1. New approach

The second approach is **Real-ESRGAN x4plus**, the first generative model in the project. It is the same RRDBNet as our Real-ESRNet baseline (23 residual-in-residual dense blocks, 16.7 M parameters), but it was trained as the generator of a GAN. We use the official pretrained weights, loaded into our own definition of the network, and run it on the same 1,360 low-resolution inputs.

## 2. Motivation

- **It isolates the effect of generative training.** Real-ESRNet and Real-ESRGAN have the same network, the same training data and the same synthetic degradations. Real-ESRGAN starts from the Real-ESRNet weights and is then trained with extra perceptual and adversarial losses. Any difference between the two is caused by the loss alone, which is the core of our research question.
- **Expected improvement.** An L1 loss averages over all plausible high-resolution images and gives smooth output. A GAN loss pushes the output towards realistic textures instead. We expect a clearly lower LPIPS (better perceptual quality) and, following the perception–distortion trade-off, a lower PSNR and SSIM than Real-ESRNet.
- **It is cheap.** The network is unchanged, so runtime and memory are the same as the baseline.

## 3. Implementation: differences from the first approach

| | Real-ESRNet (baseline) | Real-ESRGAN (new) |
|---|---|---|
| Network | RRDBNet, 16.7 M parameters | the same |
| Training data and degradations | DIV2K, Flickr2K, OST; blur, noise, resize, JPEG (quality 30–95) | the same |
| Training loss | L1 | L1 + perceptual (VGG-19) + GAN (U-Net discriminator with spectral norm) |
| Initialisation | from the ESRGAN weights | from the Real-ESRNet weights |
| Weights | `RealESRNet_x4plus.pth` | `RealESRGAN_x4plus.pth` |

- **Code changes.** `download_data.py` also fetches the Real-ESRGAN weights. `upscale.py` loads either weight file into the same `RRDBNet` (`load_rrdbnet`), so both models share one inference path: input 128×128, output 512×512, batch size 1, float32, no tiling. `evaluate.py` adds the paired difference Real-ESRGAN − Real-ESRNet.
- **Training setup.** Unchanged: nothing is trained or fine-tuned, everything runs on the dev split, and the test split is kept for the end.

## 4. Initial results and comparison with the baseline (dev split, 64 images)

PSNR and SSIM on the Y channel with a 4-pixel border crop; LPIPS with AlexNet. Mean over images. Best value per content type in bold.

| Content | Method | PSNR ↑, no JPEG | SSIM ↑ | LPIPS ↓ | PSNR ↑, JPEG 10 | SSIM ↑ | LPIPS ↓ |
|---|---|---|---|---|---|---|---|
| Objects | Bicubic | 25.62 | 0.689 | 0.488 | 22.77 | 0.543 | 0.767 |
| Objects | Real-ESRNet | **26.31** | **0.737** | 0.376 | **22.97** | **0.582** | 0.613 |
| Objects | Real-ESRGAN | 24.44 | 0.682 | **0.255** | 21.80 | 0.547 | **0.451** |
| Faces | Bicubic | 33.40 | 0.879 | 0.289 | 28.25 | 0.768 | 0.647 |
| Faces | Real-ESRNet | **33.76** | **0.894** | 0.238 | **28.72** | **0.802** | 0.456 |
| Faces | Real-ESRGAN | 31.05 | 0.859 | **0.159** | 27.65 | 0.776 | **0.342** |
| Text | Bicubic | 16.46 | 0.707 | 0.436 | 15.12 | 0.581 | 0.639 |
| Text | Real-ESRNet | **19.31** | **0.838** | 0.186 | **15.67** | **0.690** | 0.408 |
| Text | Real-ESRGAN | 17.83 | 0.818 | **0.107** | 14.38 | 0.680 | **0.212** |

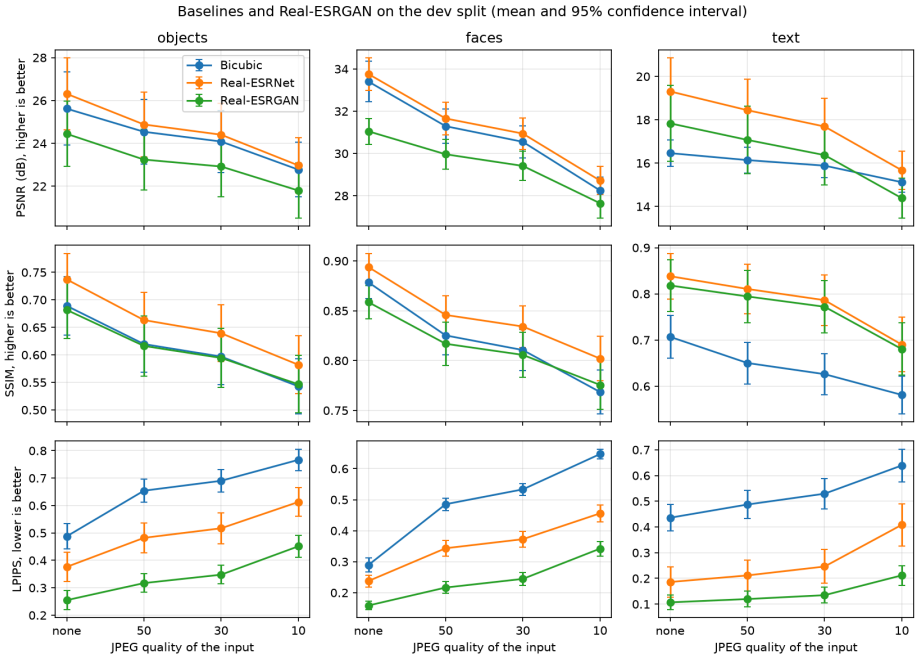
Paired difference Real-ESRGAN − Real-ESRNet on the same images, no JPEG, with 95% bootstrap interval: PSNR −1.87 ± 0.37 dB (objects), −2.71 ± 0.41 dB (faces), −1.47 ± 0.25 dB (text); LPIPS −0.12, −0.08, −0.08, that is 32%, 33% and 43% lower. Runtime on an NVIDIA RTX 3080 Ti Laptop GPU: bicubic 0.002 s, Real-ESRNet 0.11 s, Real-ESRGAN 0.11 s per image.

## 5. Problems and observations

- **The expected trade-off holds on every image.** Real-ESRGAN has a lower LPIPS than Real-ESRNet on all 256 image × JPEG pairs and a lower PSNR on 255 of them. The paired differences are outside their 95% intervals for every content type and JPEG level.
- **On small text, LPIPS rewards invented letters.** With letters 4 px and 6 px tall in the input, Real-ESRGAN is 1.8 dB and 0.5 dB *below bicubic* in PSNR, yet it has the best LPIPS. Its letters are sharp but often different from the true ones (see the text row below). An OCR error rate is needed to measure this.
- **Faces lose the most fidelity.** The PSNR drop is largest on faces (−2.7 dB): the GAN adds skin texture and sharper eyes and teeth that are not in the original. Whether this changes identity will be measured with ArcFace.
- **Under heavy compression the GAN falls below bicubic.** At JPEG 10 the gap to Real-ESRNet shrinks to 1.1–1.3 dB, but Real-ESRGAN is now under bicubic in PSNR on all three content types. On text it turns compression blocks into confident, letter-like strokes instead of smoothing them.
- **Baseline numbers reproduced on another GPU.** We re-ran all methods on an NVIDIA GPU instead of the Apple M4. Per-image values differ from Stage 2 by at most 0.003 dB PSNR and 0.0004 LPIPS; after rounding, only one entry changes (faces, bicubic LPIPS 0.290 → 0.289).

## 6. Figures

Metrics against the JPEG quality of the input (mean and 95% confidence interval, dev split):




Central 256×256 crops, input without JPEG. The text row has letters 6 px tall in the input:

